

Collatz basin densities and canonical traces from first-passage stabilization

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Abstract

We deduce from Tao’s first-passage stabilization that every fixed Collatz basin has a logarithmic density and that countable orbit labels have limits in total variation. Two incommensurable scale parameters remove the possible logarithmic oscillation left by one scale. Weighted basin densities give a finite critical Green limit at every positive integer. A first-hit time law then identifies this function with the Cesàro and Abel limits of normalized stationary Syracuse cylinder masses. For the largest cylinder mass we prove $2^{-k} \leq \max_N \pi(N + 3^k \mathbb{Z}_3) \leq (2k + 3)2^{-k}$, which determines its optimal uniform exponential rate. The canonical function is sublinear uniformly among the large ancestors of any fixed root. Its global sublinearity on nonperiodic targets is equivalent to logarithmic density one of the eventually periodic set. Additional results of Mazur upgrade the basin and label limits to natural density and imply that the canonical function is positive exactly at integers prime to three. With that input, global sublinearity is equivalent to eventual periodicity of every orbit; this condition and the exclusion of additional cycles remain unresolved. The accompanying Lean development verifies the numbered results, connecting both first-passage scales and the additional Mazur inputs to the density, clock, and canonical-limit conclusions.

1 Introduction and notation

We use Tao’s first-passage stabilization theorem to study fixed integer basins. The numbered results have accompanying Lean proofs, with the external formal inputs and their interfaces checked in one native development. We prove existence of logarithmic basin densities, total-variation limits for countable orbit labels, and equality of two canonical critical traces at every positive integer. A final proposition characterizes the global sublinearity condition that remains unresolved. We then use separate results of Mazur to obtain natural-density limits and positivity at every target prime to three.

Use the shortcut Collatz map

$$T(x) = \begin{cases} x/2, & x \text{ even}, \\ (3x + 1)/2, & x \text{ odd} \end{cases} \quad (x \in \mathbb{Z}_{>0}).$$

Write

$$\mathcal{B}_N = \{q \geq 1 : T^j(q) = N \text{ for some } j \geq 0\}, \quad H_E(X) = \sum_{\substack{q \leq X \\ q \in E}} \frac{1}{q}.$$

The logarithmic density of E , when it exists, is $d_{\log}(E) = \lim_{X \rightarrow \infty} H_E(X)/\log X$. For odd sets we also use relative logarithmic density, whose denominator is

$$H_{\text{odd}}(X) = \frac{1}{2} \log X + O(1).$$

A target is *periodic* if it lies on a finite cycle; a nonperiodic target can still be eventually periodic. A *divergent orbit* never enters a cycle. Its integer states are distinct and tend to infinity. Two positive integers belong to the same *tail component* when their forward orbits eventually meet.

1.1 Relation to the 2026 descent and formalization results

The density, target, time, and observable must be distinguished when comparing the present statements with recent work. Shaik’s September revision [3], version 4.0.0, proves natural-density descent to

$$(\log n)^{A_{\text{FP}}} U(n), \quad A_{\text{FP}} = \frac{1}{2(1 - H_2(\log_3 2))},$$

where H_2 is binary entropy, for every prescribed positive $U(n) \rightarrow \infty$. The shortcut and raw Collatz witnesses have a common landing, clocks centered at $2 \log n / \lambda$ and $3 \log n / \lambda$, respectively, with $O(\sqrt{\log n \log \log n})$ errors, and a common height bound $nR(n)$ for every prescribed positive $R(n) \rightarrow \infty$. His fixed-exponent specialization with a balanced height ceiling attains the joint exceptional exponent $(1 - H_2(\log_3 2))A - 1/2$. Our fixed-target first-hit laws concern membership in a specified basin and do not claim new drift constants. The target in Shaik’s theorem still tends to infinity; that theorem does not itself supply a limit for the density of a specified basin.

Nashida’s fixed-barrier preprint [4] proves

$$\#\{n \leq X : \text{Col}_{\min}(n) > M\} \leq KXM^{-c'}, \quad c' \simeq 3.506 \cdot 10^{-6},$$

uniformly in the endpoint X and integer barrier $M \geq 1$. Its proof combines fixed-barrier counting assembled from Shaik’s formalization with Mazur’s natural-density recursion. This is a stronger quantitative tightness estimate than is needed below; together with our existence of the minimum law it gives $\mu_{\min}((M, \infty)) \leq KM^{-c'}$. It neither identifies a specified basin’s mass nor makes a zero-density exceptional set empty. Nashida’s generalized-map preprint [5] proves the analogous power saving for the Gonçalves–Greenfeld–Madrid family. His separate stopping-time preprint [6] improves the Terras-type exponent for long stopping times. That count ranges over starting integers and is distinct from the uniform packing bound for one finite trajectory of distinct states used here.

Mazur’s formalization of Tao [7] includes the first-passage stabilization theorem, rather than only Tao’s final almost-boundedness conclusion. Its checked source fixes the scale parameter at $\alpha = 1001/1000$. We rebuilt that theorem and adapted its checked source to $\alpha = 2001/2000$, retaining the real-threshold first-passage and rate conclusions. The two checked scale namespaces coexist in the combined Lean development and supply exactly the scales used in Proposition 2; they do not assert a formally checked continuum of scale parameters. His natural-density transport theorem [8] compares uniform and logarithmic first-passage laws on the same source block. Section 12 uses that additional input to upgrade our basin and label limits to natural density. His predecessor theorem [9] also implies positivity of the canonical trace at every integer prime to three. These are credited consequences of those external results, not claims that the external manuscripts state our weighted or canonical-limit theorems.

The formal verification boundaries differ between these sources. Shaik’s version-4 theorem map distinguishes its checked principal chain from additional manuscript-only refinements; the separate Palomar version-3 record concerns the preceding revision. Nashida’s fixed-barrier theorem has no additional mathematical Lean premise. His generalized natural-density theorem takes a transcription of Matveev’s bound as a hypothesis, and his stopping-time improvement takes a transcription of Bugeaud’s bound; the cited published theorems supply those inputs to the written proofs. The explicit numerical saving in the latter paper is certified by rational interval calculations outside Lean. Our inspection of those theorem statements and source records does not constitute a local rebuild of their full formalizations.

2 Uniform control of finite distinct trajectories

For a finite shortcut path $x_0, \dots, x_\ell$, define its correction product, excluding the terminal state, by

$$P(x_0, \dots, x_\ell) = \prod_{\substack{0 \leq i < \ell \\ x_i \text{ odd}}} \left(1 + \frac{1}{3x_i}\right).$$

If the path has r odd steps, then

$$x_\ell = \frac{3^r}{2^\ell} x_0 P(x_0, \dots, x_\ell). \quad (1)$$

The bound below is uniform in the length and endpoints of a distinct path.

Lemma 1. *Put*

$$\beta = H_2(\log_3 2) < 1, \quad H_2(t) = -t \log_2 t - (1-t) \log_2(1-t).$$

For every $b \in (\beta, 1)$, there are constants $K_b, R_b < \infty$ such that every finite path of distinct positive states satisfies

$$\#\{i : x_i \leq X\} \leq K_b X^b \quad (X \geq 1), \quad (2)$$

$$\sum_{\substack{i \\ x_i \geq M}} \frac{1}{x_i} \leq R_b M^{b-1} \quad (M \geq 1). \quad (3)$$

In particular there is a constant $P_0 < \infty$, independent of the path, for which $1 \leq P \leq P_0$.

Proof. Choose $\rho \in (1/2, \log_3 2)$ with $H_2(\rho) = b$, and put $c = \rho \log_2 3 < 1$. Let $A(X) = \#\{i : x_i \leq X\}$ and $h = \lfloor \log_2 X \rfloor$. Length- h parity words correspond bijectively to residue classes modulo 2^h , by successive determination of the parity at each step. A word with r odd positions has affine slope $3^r/2^h$ and offset at most $(3/2)^r - 1$. Hence a start at most X with $r \leq \rho h$ has its h -step image below $3 \cdot 3^{\rho h} \leq 3X^c$.

Images whose complete h -step windows lie in the segment are distinct. At most h indices are too close to its terminal endpoint. The elementary binomial entropy bound gives at most $2^{hH_2(\rho)}$ words with more than ρh odd positions, and each residue class contains at most two possible starts in $[1, X]$. Consequently

$$A(X) \leq A(3X^c) + 2X^b + \lfloor \log_2 X \rfloor.$$

This also holds if the segment has length less than h . Choose X_0 so large that $3^b X^{-b(1-c)} \leq 1/2$ and $3X^c < X$ for $X \geq X_0$. Since $\log_2 X \leq X^b/(be \log 2)$, strong induction on integer X , using $A(X) \leq X$ in the initial range, proves (2) for a sufficiently large constant K_b . Taking integer parts handles real X .

Divide the states at least M into the shells $[2^j M, 2^{j+1} M)$. Equation (2) gives

$$\sum_{x_i \geq M} \frac{1}{x_i} \leq K_b 2^b M^{b-1} \sum_{j \geq 0} 2^{j(b-1)} =: R_b M^{b-1}.$$

Finally, $\log P \leq \frac{1}{3} \sum_i 1/x_i$, so one may take $P_0 = \exp(R_b/3)$. $\square$

The same tail estimate applies to any subset of the states. In particular, if the odd source states of a distinct path all exceed M , its correction product is

$$1 \leq P \leq \exp\left(\frac{R_b}{3} M^{b-1}\right). \quad (4)$$

The intervening even states need not exceed M .

3 The first-passage input at two scales

On positive odd integers let

$$S(q) = \frac{3q+1}{2^{v_2(3q+1)}}, \quad \lambda = \log(4/3), \quad \delta = \frac{\lambda}{2}.$$

Let $\tau_x(q)$ be the first S -time at a value at most x , with value infinity on failure. Write $\text{Pass}_x(q)$ for that landing value, and give it the default value 1 on failure. For $a > 1$, let $Q_y^{(a)}$ have logarithmic distribution on the odd integers in $[y, y^a]$. The default convention ensures the exact identity

$$\text{Pass}_M \circ \text{Pass}_x = \text{Pass}_M \quad (1 \leq M \leq x), \quad (5)$$

including trajectories on which passage fails.

Proposition 2 (Tao's stabilization at the required scales). *For each*

$$a \in \left\{ \frac{1001}{1000}, \frac{2001}{2000} \right\}$$

there are $C, c > 0$ such that, for all sufficiently large x ,

$$\Pr(\tau_x(Q_y^{(a)}) = \infty) \leq Cx^{-c}, \quad y = x^a, \quad x^{a^2}, \quad (6)$$

$$d_{\text{TV}}(\mathcal{L}(\text{Pass}_x(Q_{x^a}^{(a)})), \mathcal{L}(\text{Pass}_x(Q_{x^{a^2}}^{(a)}))) \leq C(\log x)^{-c}. \quad (7)$$

Proof. For $a = 1.001$, these are [1, Proposition 1.11, equations (1.19)–(1.20)]. We record the parameter check needed to use a second scale. In the proof in [1, Section 5], retain

$$n_0 = \left\lfloor \frac{\log x}{10 \log 2} \right\rfloor, \quad m_0 = \left\lfloor \frac{a-1}{100} \log x \right\rfloor.$$

The input Propositions 1.9 and 1.14 there are independent of a . All numerical margins in Section 5 remain strict, uniformly on the compact interval $[1.0005, 1.001]$ containing both scales. Reduction of either source modulo 2^{3n_0} has error $O(x^{0.3-a}/\log x)$, smaller than the required $x^{-0.3}$. The short-time upper exponent used after equation (5.5) is at most

$$a^3 + \frac{\log_2 3 - 1.9}{10} < 0.971501 < 0.99.$$

The passage window in equation (5.9) has endpoints of order $(a-1)\log x/\lambda$ and $(a^3-1)\log x/\lambda$. Its lower endpoint is much larger than $2m_0$, and its upper endpoint is below $0.01044 \log x < n_0$. The buffers $(\log x)^{0.6}$, $(\log x)^{0.7}$, $(\log x)^{0.8}$ are smaller than these linear margins. The affine offset is at most $3^{n_0} = O(x^{(\log_2 3)/10})$, whereas the main term in the regular-valuation regime is at least

$$x^{a+(\log_2(3/4))/10-o(1)}.$$

Thus the relative offset estimates used in equations (5.13)–(5.19) also retain their margins.

For completeness, the residue summation in Lemma 5.3 can be bounded uniformly without discarding the factor a_{m_0} . With $A = a_1 + \dots + a_{m_0}$, the ordinary contribution is bounded by a constant times $2^{-A}a_{m_0}$, whose sum over all positive tuples is 2. The additional large-total contribution is at most

$$3^n \Pr(G_1 + \dots + G_{m_0} > \tfrac{1}{2} \log_2 x), \quad \Pr(G_i = j) = 2^{-j}.$$

Exponential Markov at parameter $(\log 2)/2$ bounds it by

$$x^{(\log_2 3)/10-1/4} (1 + \sqrt{2})^{m_0} = O(x^{-\eta})$$

for some $\eta > 0$, since $m_0 \leq 10^{-5} \log x$. Finally $m_0 \asymp \log x$, the passage-window count and source normalization have the same factor $(a-1) \log y$, and $n - m_0 \geq m_0$. The application of Proposition 1.14 at depth m_0 , and the remaining concentration errors in that proof, therefore give the same negative power of $\log x$. This verifies both required scales by the published proof; it is not an additional hypothesis about its printed proposition. The accompanying replay of Mazur's Lean sources checks the two scales separately. Its source law is the reciprocal-weighted law on the inclusive odd interval $[y, y^a]$, and its first-passage location uses the same default value 1 as above. Its distance is full ℓ^1 , so its bound implies (7) with our total-variation convention. $\square$

We use

$$a = \frac{1001}{1000}, \quad b = \frac{2001}{2000}. \quad (8)$$

Their logarithms have irrational ratio: equality of positive integer powers would contradict the exponent of the prime 7.

Lemma 3 (Two-scale averaging). *Let V be a Banach space and $A : [t_0, \infty) \rightarrow V$ satisfy*

$$\|A(t)\| \leq Lt + C, \quad \|A(t+h) - A(t)\| \leq Lh + C \quad (h \geq 0).$$

For $u > 1$, put

$$p_u(t) = \frac{A(ut) - A(t)}{(u-1)t}.$$

Suppose $\log a / \log b \notin \mathbb{Q}$ and, uniformly as $t \rightarrow \infty$,

$$\|p_a(at) - p_a(t)\| = O(t^{-c}) \quad (c > 0), \quad \|p_b(bt) - p_b(t)\| = o(1).$$

Then $A(t)/t$ converges in norm. Its error is $O(t^{-\eta})$ for every $0 < \eta < \min\{c, 1\}$.

Proof. Set $B(u) = e^{-u}A(e^u)$ and $\ell = \log a$. The increment bounds imply boundedness of B and

$$\|B(v) - B(u)\| \leq 2L(v-u) + O(e^{-u}) \quad (v \geq u).$$

For

$$Q(u) = \frac{aB(u+\ell) - B(u)}{a-1}$$

the first scale estimate reads $\|Q(u+\ell) - Q(u)\| = O(e^{-cu})$. The geometric tail is summable, so completeness gives

$$P(u) = \lim_{k \rightarrow \infty} Q(u+k\ell), \quad P(u+\ell) = P(u), \quad Q(u) - P(u) = O(e^{-cu}).$$

Extend P periodically to all real u . The exact recurrence

$$(B-P)(u+\ell) = a^{-1}(B-P)(u) + (1-a^{-1})(Q-P)(u)$$

shows, by iteration from one fixed interval of length ℓ , that $B(u) - P(u) = O(e^{-\eta u})$ for every $\eta < \min\{c, 1\}$. The modulus for B , shifted by $k\ell$ and then passed to the limit, makes P Lipschitz.

Put $m = \log b$. The second scale estimate gives

$$bB(u+2m) - (b+1)B(u+m) + B(u) \rightarrow 0.$$

Replace u by $u+k\ell$ and use the preceding convergence to obtain the same expression with P equal to zero. For $D(u) = P(u+m) - P(u)$, this says $bD(u+m) = D(u)$. Backward iteration and boundedness force $D = 0$. Thus P has the two incommensurable periods ℓ, m ; continuity makes it constant. The asserted convergence and rate follow. $\square$

4 Elementary local passage estimates

The following estimates concern the full unstopped valuation prefix. They do not assume independence after a hitting time.

Lemma 4. *Uniformly for $a \in \{1001/1000, 2001/2000\}$, let Q have logarithmic distribution on the odd integers in either $[x^a, x^{a^2}]$ or $[x^{a^2}, x^{a^3}]$. There are $C, c > 0$ such that*

$$\Pr\left(\tau_x(Q) = \infty \text{ or } \left|\tau_x(Q) - \frac{\log(Q/x)}{\lambda}\right| > C(\log x)^{3/5}\right) \leq Ce^{-c(\log x)^{1/5}}, \quad (9)$$

$$\Pr(\text{Pass}_x(Q) \leq \sqrt{x}) \leq Cx^{-c}, \quad (10)$$

$$\sup_{m \leq x} \Pr(\text{Pass}_x(Q) = m) \leq Cx^{-c}. \quad (11)$$

The last two estimates include the default atom on passage failure.

Proof. Put

$$R = \lfloor 0.1 \log x \rfloor, \quad B = \lfloor \tfrac{1}{2} \log_2 x \rfloor, \quad E = (\log x)^{3/5}.$$

For the actual trajectory let

$$a_i = v_2(3S^{i-1}(Q) + 1), \quad A_k = \sum_{i=1}^k a_i.$$

Reduction of either source law modulo 2^{B+1} differs in total variation from the uniform odd-residue law by

$$O\left(\frac{2^B}{x^a \log x}\right). \quad (12)$$

Indeed the reciprocal sum in each residue class differs from its integral by $O(x^{-a})$, while the source harmonic mass is a positive constant times $\log x$, uniformly in the stated range. A prescribed positive valuation tuple of total $A_R \leq B$ occupies one odd residue class modulo 2^{A_R+1} ; under the uniform law its probability is 2^{-A_R} . These tuples are disjoint. Their probabilities, and that of their complement, are therefore the same as for independent geometric variables $\Pr(G_i = j) = 2^{-j}$, up to (12).

The moment generating function

$$\mathbb{E}e^{tG_i} = \frac{e^t}{2 - e^t} \quad (t < \log 2)$$

gives

$$\Pr\left(\left|\sum_{i=1}^k G_i - 2k\right| > E\right) \leq 2 \exp(-c \min\{E^2/k, E\}).$$

A union bound for $k \leq R$, and $B/R \rightarrow 5/\log 2 > 7$, show that, outside probability $Ce^{-c(\log x)^{1/5}}$,

$$A_R \leq B, \quad |A_k - 2k| \leq E \quad (0 \leq k \leq R). \quad (13)$$

Here the geometric upper tail $A_R > B$ and the residue error are polynomially small in x .

The exact affine formula is

$$S^k(Q) = 3^k 2^{-A_k} Q + F_k, \quad 0 \leq F_k \leq 3^k.$$

On (13), its main term is at least $x^a(3/4)^R 2^{-E}$, so the relative offset is at most

$$x^{0.1 \log 4 - a} 2^E \leq x^{-4/5}$$

for sufficiently large x . Thus, uniformly for $k \leq R$,

$$\left| \log S^k(Q) - (\log Q - \lambda k) \right| \leq E. \quad (14)$$

The central crossing time satisfies

$$\frac{a-1}{\lambda} \log x \leq \frac{\log(Q/x)}{\lambda} \leq \frac{a^3-1}{\lambda} \log x < 0.01044 \log x.$$

Bracketing it by an error $2E/\lambda + 1$ stays inside $[0, R]$. Equation (14) rules out an earlier crossing and guarantees a crossing by the upper endpoint. This proves (9). It also bounds the landing below by $x \exp(-3E - \lambda)$.

For the polynomial estimates, we use a less precise good event. If $\tau_x(Q) > R$, the product identity, with all its first R odd source states exceeding x , implies

$$A_R \log 2 < (a^3 - 1) \log x + R \log 3 + \frac{R}{3x} \leq 1.8R \log 2$$

for large x . The limiting upper bound for A_R/R is at most

$$\log_2 3 + \frac{1.001^3 - 1}{0.1 \log 2} < 1.63.$$

Since $B \geq 6R$ eventually, the geometric lower and upper tails and (12) give

$$\Pr(\tau_x(Q) > R) + \Pr(A_R > B) \leq Cx^{-c}. \quad (15)$$

On the complementary event, a landing at most $\sqrt{x}$ would require the last valuation, whose source is greater than x , to exceed $\frac{1}{2} \log_2 x + \log_2 3 > B$. This proves (10), including failure.

Finally fix a landing value m . On the event $\tau_x(Q) \leq R$, $A_R \leq B$, its stopped positive valuation tuple has total at most B . The affine formula shows that this tuple and m determine at most one starting integer. There are $\sum_{A=1}^B 2^{A-1} < 2^B$ such tuples, allowing all lengths. Each source atom has probability $O(x^{-a}/\log x)$. Adding (15) proves (11). $\square$

5 Logarithmic laws for basins and countable labels

Proposition 5. *Let I be countable and let F assign a label in I to each positive odd integer. Suppose that, for all sufficiently large x , successful first passage preserves its label:*

$$F(q) = F(\text{Pass}_x(q)) \quad \text{whenever } \tau_x(q) < \infty.$$

Then the law of F under logarithmically uniform odd starts in $[1, X]$ converges in total variation to a probability on I . This applies to tail components, global orbit minima, and every fixed first-passage location. Their limits also hold for all positive starts, with the same probabilities, after applying F to the odd part.

Proof. In $V = \ell^1(I)$, define

$$A_F(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{\delta_{F(q)}}{q}.$$

Its norm is $t/2 + O(1)$, and its increments have norm at most $h/2 + O(e^{-t})$. Push (7) forward under F ; the discrepancy between the original and first-passage labels is supported on passage failure, controlled by (6). The harmonic mass of the odd interval $[e^t, e^{at}]$ is $(a-1)t/2 + O(e^{-t})$. With $t = a \log x$, normalization gives

$$\|p_a(at) - p_a(t)\|_1 = O(t^{-c})$$

for each of the two scales in (8). Lemma 3 gives norm convergence of $A_F(t)/t$. The limiting vector is nonnegative and has norm $1/2$; after multiplication by two it is a probability. Norm convergence gives total-variation convergence, uniformly over every subset of labels. In particular there is no loss of mass among countably many labels.

For a component label, successful passage stays in the same component. For the global orbit minimum, every odd state before passage exceeds x , while the landing is at most x ; an even intermediate state is followed by an odd state no larger than itself. The minimum is therefore preserved. For the label $F = \text{Pass}_M$ at fixed M , use (5) for $x \geq M$.

These labels, extended by taking the odd part, are invariant under doubling. Write a positive start as $2^h q$, q odd. For any fixed cutoff H , the corresponding harmonic sum of label vectors is

$$\sum_{h=0}^H 2^{-h} A_F(\log X - h \log 2).$$

The omitted starts are multiples of 2^{H+1} , with logarithmic proportion tending to 2^{-H-1} . First pass $X \rightarrow \infty$, then $H \rightarrow \infty$, to obtain the same limiting probability. $\square$

Let μ_{tail} and $\mu_{\min}$ denote the component and minimum probabilities. In particular every union of tail components has a logarithmic density, and

$$d_{\log}\{q : \min_{j \geq 0} T^j(q) > M\} = \mu_{\min}((M, \infty)) \rightarrow 0. \quad (16)$$

For fixed M , let Λ_M denote the limiting law of Pass_M on odd starts. Equation (5) implies

$$(\text{Pass}_L)_* \Lambda_M = \Lambda_L \quad (L \leq M).$$

These probabilities do not determine the mass of any specified component or the probability of convergence to one.

Proposition 6. *Every fixed shortcut basin $\mathcal{B}_N$ has a logarithmic density.*

Proof. On successful first passage with $x \geq N$, its membership can change only if the even target N is visited during the last halving chain and then omitted by the Syracuse acceleration. That event has landing $\text{oddpart}(N)$, and for $x \geq N^2$ it is included in $\text{Pass}_x \leq \sqrt{x}$. Thus for both source laws in Proposition 2, the expectations of $\mathbf{1}_{\mathcal{B}_N}(Q)$ and $\mathbf{1}_{\mathcal{B}_N}(\text{Pass}_x(Q))$ differ by $O(x^{-c})$, using failure and (10). For an odd target there is no skipped crossing. This argument also covers the default value for a periodic target such as one.

Applying (7) to the bounded indicator gives the two-scale estimates for

$$A_N(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}, q \in \mathcal{B}_N}} \frac{1}{q}.$$

Lemma 3 proves convergence of $2A_N(t)/t$. For a start $2^h q$, membership agrees with that of its odd part unless its initial halving chain hits N . These exceptional starts lie on the single ray $\{2^h \text{oddpart}(N) : h \geq 0\}$, of finite reciprocal sum. The finite-valuation decomposition in the preceding proof therefore gives the all-positive density with the same value. $\square$

6 Weighted densities and the Green limit

For $q \in \mathcal{B}_N$, let $P_N(q)$ be the correction product on the path to its first visit to N , and define

$$w_N(q) = P_N(q)^{-1} \quad (q \in \mathcal{B}_N), \quad w_N(q) = 0 \quad (q \notin \mathcal{B}_N).$$

The first-hit path is distinct, even when N is periodic. Lemma 1 gives

$$P_0^{-1} \mathbf{1}_{\mathcal{B}_N} \leq w_N \leq \mathbf{1}_{\mathcal{B}_N}. \quad (17)$$

Proposition 7. *For every $N \geq 1$, the limit*

$$D_N = \lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{q \leq X} \frac{w_N(q)}{q}$$

exists. The odd harmonic sum has limit $D_N/2$, and

$$P_0^{-1} d_{\log}(\mathcal{B}_N) \leq D_N \leq d_{\log}(\mathcal{B}_N). \quad (18)$$

Proof. Fix a packing exponent $\gamma \in (\beta, 1)$. On successful passage below x , outside a skipped even target, first-hit paths concatenate:

$$w_N(q) = \frac{w_N(\text{Pass}_x(q))}{P_{\text{prefix}}}.$$

This identity includes the case where both weights are zero. The prefix is distinct and its odd source states exceed x . By (4),

$$|w_N(q) - w_N(\text{Pass}_x(q))| \leq Cx^{\gamma-1}.$$

Use the bound one on failure or a skipped crossing. For the two source laws, their probabilities are $O(x^{-c})$, as in Proposition 6. Hence, for fixed N and all sufficiently large x ,

$$|\mathbb{E}w_N(Q) - \mathbb{E}w_N(\text{Pass}_x(Q))| \leq Cx^{\gamma-1} + Cx^{-c}.$$

Combined with (7), this proves the two-scale estimates for

$$A(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{w_N(q)}{q}.$$

This nonnegative scalar function satisfies the increment bounds in Lemma 3, which therefore gives its limiting mean.

Initial halvings contribute no correction factor. Thus $w_N(2^h q) = w_N(q)$ outside the same single exceptional ray as in the preceding proof. Its reciprocal sum is finite. The finite-valuation decomposition gives the full mean D_N and the factor $1/2$ for the odd mean. Finally use (17). $\square$

For $1 < s < 2$, define the positive inverse operator and its Green sum by

$$(\mathcal{R}_s f)(v) = 2^{-s} f(2v) + (3/2)^s \mathbf{1}_{v \equiv 2 \pmod{3}} f((2v-1)/3), \quad (19)$$

$$M_s(v) = \sum_{j \geq 0} \mathcal{R}_s^j \mathbf{1}(v), \quad \kappa_s = 2^{-s} + \frac{(3/2)^s}{3}, \quad C_s(v) = (1 - \kappa_s) M_s(v). \quad (20)$$

Strict convexity gives $\kappa_s < 1$ on $1 < s < 2$, and

$$1 - \kappa_s = \delta(s-1) + O((s-1)^2). \quad (21)$$

If N is periodic, let ℓ_N be its shortcut period, r_N its number of odd steps, and $\beta_N = 3^{r_N}/2^{\ell_N}$. The return map has a strictly positive affine offset, so $0 < \beta_N < 1$. Put

$$B_N = \begin{cases} 1, & N \text{ nonperiodic,} \\ (1 - \beta_N)^{-1}, & N \text{ periodic.} \end{cases}$$

This scalar B_N is distinct from the basin $\mathcal{B}_N$.

Theorem 8. *At every positive integer the Green sum is finite for $1 < s < 2$, and its critical limit exists:*

$$g(N) := \lim_{s \downarrow 1} C_s(N) = \delta B_N N D_N. \quad (22)$$

It is a finite nonnegative function satisfying

$$g(v) = \frac{1}{2}g(2v) + \frac{3}{2}\mathbf{1}_{v \equiv 2 \pmod{3}}g((2v-1)/3). \quad (23)$$

Moreover $g(N) > 0$ if and only if $d_{\log}(\mathcal{B}_N) > 0$.

Proof. The inverse word corresponding to the first hit from q has weight $(N/(qP_N(q)))^s$, by (1). At a nonperiodic target no endpoint can hit it at two different times. At a periodic target all later hits are obtained by complete cycle traversals. Consequently

$$M_s(N) = \begin{cases} N^s \sum_{\substack{q \geq 1 \\ q \mid N}} w_N(q)^s q^{-s}, & N \text{ nonperiodic,} \\ \frac{N^s}{1 - \beta_N^s} \sum_{q \geq 1} w_N(q)^s q^{-s}, & N \text{ periodic.} \end{cases} \quad (24)$$

This proves finiteness by comparison with $\zeta(s)$.

Write $A(t) = \sum_{q \leq e^t} w_N(q)/q = D_N t + o(t)$. For $\varepsilon = s - 1 > 0$, Stieltjes integration by parts gives

$$\varepsilon \sum_{q \geq 1} w_N(q) q^{-1-\varepsilon} = \varepsilon^2 \int_0^\infty e^{-\varepsilon t} A(t) dt \longrightarrow D_N.$$

The convention at zero includes a possible atom $q = 1$. The bound $A(t) \leq t + O(1)$ justifies the change of variables and dominated convergence. For $0 \leq w \leq 1$,

$$0 \leq w - w^s = \int_1^s (-w^r \log w) dr \leq \frac{s-1}{e}.$$

Thus replacing w_N by w_N^s in the preceding Abelian limit changes it by at most $(s-1)^2 \zeta(s)/e \rightarrow 0$. Equations (21) and (24) prove (22).

The resolvent equation is $M_s = 1 + \mathcal{R}_s M_s$. It has only two inverse branches at any fixed integer, so passage to the finite pointwise limits gives (23). The positivity equivalence follows from (18). $\square$

7 Uniform bounds for backward basins

Proposition 9. *There is a function $\varepsilon(M) = O((\log M)^{-c}) \rightarrow 0$ such that*

$$\mathcal{B}_v \cap [1, M] = \emptyset \implies d_{\log}(\mathcal{B}_v) \leq \varepsilon(M). \quad (25)$$

Consequently, for every fixed positive root n ,

$$\lim_{\substack{v \rightarrow \infty \\ v \in \mathcal{B}_n}} d_{\log}(\mathcal{B}_v) = 0, \quad \lim_{\substack{v \rightarrow \infty \\ v \in \mathcal{B}_n}} \frac{g(v)}{v} = 0. \quad (26)$$

Both limits are uniform over the indicated targets.

Proof. Fix $a = 1.001$, suppress its superscript on Q_y , and write

$$\nu_x = \mathcal{L}(\text{Pass}_x(Q_{x^a})), \quad x_j = M^{a^j}.$$

Pushing (7) through Pass_M , and using (5), gives

$$d_{\text{TV}}((\text{Pass}_M)_*\nu_{x^a}, (\text{Pass}_M)_*\nu_x) \leq C(\log x)^{-c} \quad (x \geq M).$$

The sum along x_j is at most $C(\log M)^{-c}/(1 - a^{-c})$. Combining it with the local atom estimate (11) at M proves, uniformly in j and $m \leq M$,

$$\Pr(\text{Pass}_M(Q_{x_j^a}) = m) \leq CM^{-c} + C'(\log M)^{-c} =: \eta(M).$$

For $E_{M,m} = \{q \text{ odd} : \text{Pass}_M(q) = m\}$, the source intervals $[x_j^a, x_j^{a^2}]$ tile $[M^a, \infty)$, apart from shared endpoints. A final partial interval ending at X is contained in its full interval, whose right endpoint is at most X^a . Shared endpoints have finite total reciprocal mass. Hence

$$H_{E_{M,m}}(X) \leq \eta(M)H_{\text{odd}}(X^a) + O_M(1)$$

with an error independent of m , and the relative odd upper logarithmic density is at most $a\eta(M)$.

Now suppose $\mathcal{B}_v \cap [1, M] = \emptyset$, let $u = \text{oddpart}(v)$, and put $m_v = \text{Pass}_M(u)$. The value $m_v = 1$ is allowed when passage fails. Every odd start reaching v has no state at most M before that hit. Its first odd landing at most M is consequently the same as that of u , or both passages fail. For an arbitrary start whose initial halving chain already reaches v , its odd part is exactly u . Therefore

$$\mathcal{B}_v \subseteq \{2^h q : h \geq 0, q \in E_{M,m_v}\}.$$

The dyadic closure of an odd set of relative upper logarithmic density at most d has ordinary upper logarithmic density at most d : sum finitely many valuations and bound the omitted starts by multiples of 2^{H+1} , then let $H \rightarrow \infty$. This proves (25) with $\varepsilon(M) = a\eta(M)$.

For fixed q, n , the intersection of the forward orbit of q with $\mathcal{B}_n$ is finite. If nonempty, that orbit reaches n ; its prefix is finite, and a later return to $\mathcal{B}_n$ would make n periodic, leaving only one finite cycle. Thus

$$F_{M,n} = \bigcup_{q \leq M} (\{T^j(q) : j \geq 0\} \cap \mathcal{B}_n)$$

is finite. Above its maximum, every $v \in \mathcal{B}_n$ has no ancestor at most M , so (25) gives the first limit in (26). Only finitely many targets in $\mathcal{B}_n$ can be periodic: any such target and n lie on the same finite cycle. For all the remaining targets,

$$\frac{g(v)}{v} = \delta D_v \leq \delta d_{\log}(\mathcal{B}_v).$$

This proves the second limit. □

8 A first-hit law for the number of odd steps

For $q \in \mathcal{B}_N$, let $k_N(q)$ count the odd source states before the first shortcut visit to N . For odd q, N this is the first Syracuse hitting time. For an even target it includes the last odd step even when the hit occurs before the end of its halving chain.

Proposition 10. *For every fixed $N \geq 1$ and $\epsilon > 0$, the odd set*

$$\left\{ q \in \mathcal{B}_N, q \text{ odd} : \left| k_N(q) - \frac{\log q}{\lambda} \right| > \frac{\epsilon \log q}{\lambda} \right\} \quad (27)$$

has logarithmic density zero.

Proof. Again fix $a = 1.001$. On the same reference source Q_{x^a} , apply (9) at thresholds x and $y = x^{1/a}$. Its interval is $[x^a, x^{a^2}] = [y^{a^2}, y^{a^3}]$, so both local estimates apply. With probability at least $1 - Ce^{-c(\log x)^{1/5}}$, both passages succeed and

$$\tau_y(Q_{x^a}) - \tau_x(Q_{x^a}) = \frac{\log x - \log y}{\lambda} + O((\log x)^{3/5}).$$

On this event the difference is exactly $\tau_y(\text{Pass}_x(Q_{x^a}))$. Thus the displayed stage estimate is an event about a single landing variable with law ν_x .

Fix a large bottom threshold M , put $x_i = M^{a^i}$, and start from $Q = Q_{x_j^a}$. Telescoping (7) and using (5) give

$$d_{\text{TV}}(\mathcal{L}(\text{Pass}_{x_i}(Q)), \nu_{x_i}) \leq C \sum_{r=i}^{j-1} (\log x_r)^{-c} \leq C' (\log x_i)^{-c} \quad (i \leq j). \quad (28)$$

At $i = j$ the laws agree. A union bound for the stage events, transferred under (28), together with the local top passage estimate, has total error at most

$$Ce^{-c(\log x_j)^{1/5}} + \sum_{i=1}^j \left(Ce^{-c(\log x_{i-1})^{1/5}} + C' (\log x_i)^{-c} \right) \leq C'' (\log M)^{-c'}$$

uniformly in j . Only marginal total variation is needed, since each stage event is a function of its own landing variable. No independence between stages is asserted.

On the simultaneous good event, top success and success of every stage permit exact time concatenation. The main terms telescope, and

$$\sum_{i=0}^j (\log x_i)^{3/5} \leq C_a (\log Q)^{3/5}.$$

It follows that

$$\Pr \left(\tau_M(Q) = \infty \text{ or } \left| \tau_M(Q) - \frac{\log Q - \log M}{\lambda} \right| > C_a (\log Q)^{3/5} \right) \leq C'' (\log M)^{-c'}. \quad (29)$$

The default-valued identities were used only to compare landing laws; actual time concatenation is performed on the successful event.

Take $M \geq N$. Among the finitely many odd $u \leq M$ in $\mathcal{B}_N$, let $R_{M,N}$ be the maximum of $k_N(u)$, or zero if there are none. For an odd $q > M$ in $\mathcal{B}_N$, passage below M must succeed. If it occurs before the first visit to N , counts concatenate. If an even N is first visited inside the final halving chain of that passage, its odd-step count is already exactly $\tau_M(q)$. Therefore

$$0 \leq k_N(q) - \tau_M(q) \leq R_{M,N}. \quad (30)$$

For fixed M, N, ϵ , the bounded terms $R_{M,N} + (\log M)/\lambda$ and the $O((\log q)^{3/5})$ error are smaller than $\epsilon \log q / \lambda$ on all sufficiently high source intervals. By (29), the probability of (27) on each such interval is at most $C'' (\log M)^{-c'}$.

The intervals $[x_j^a, x_j^{a^2}]$ tile $[M^a, \infty)$. As in Proposition 9, covering the last partial interval costs at most the multiplicative factor a in logarithmic mass. Thus the odd-relative upper density of (27) is at most $aC'' (\log M)^{-c'}$. The set does not depend on M ; letting $M \rightarrow \infty$ proves the claim. $\square$

The conclusion is an absolute zero-density statement, also when $\mathcal{B}_N$ has density zero. Multiplying its harmonic summands by any weights in $[0, 1]$ preserves it. No bound uniform over all starting integers or all targets is asserted.

The corresponding lower-clock constraint is also consistent with Inslemann's natural-density trajectory estimate [2, Theorem 1.10]. The two-sided fixed-target law above uses the transported upper-time estimate as well.

Proposition 11 (Conversion of the first-hit clocks). *Let $\ell_N(q)$ be the shortcut first hitting time of N . For every fixed N , in logarithmic density on all positive starts in $\mathcal{B}_N$,*

$$k_N(q) = \frac{\log q}{\lambda} + o(\log q), \quad \ell_N(q) = \frac{2 \log q}{\lambda} + o(\log q).$$

Precisely, for each positive error tolerance the set violating either corresponding relative error bound has logarithmic density zero.

Proof. Write $q = 2^h u$ with u odd. Outside the single ray whose initial halving chain hits N , one has $k_N(q) = k_N(u)$ and $\ell_N(q) = h + \ell_N(u)$. The ray has finite reciprocal sum. For every fixed h , $\log q - \log u = h \log 2$ is bounded; hence Proposition 10 gives the odd-clock conclusion on that valuation class. Finitely many such classes suffice, because the remaining starts have upper logarithmic density at most 2^{-H-1} , which tends to zero with the cutoff.

The exact first-hit product formula is

$$\ell_N(q) \log 2 = k_N(q) \log 3 + \log q - \log N + \log P_N(q).$$

Since $1 \leq P_N(q) \leq P_0$, its last two terms beyond $\log q$ are bounded for fixed N . The identity $\log 3 + \lambda = \log 4 = 2 \log 2$ gives the shortcut-clock coefficient. $\square$

9 Stationary cylinders and their maximal mass

We define the stationary measure explicitly so that its values at ordinary integers require no choice of an almost-everywhere density. Let $G_1, G_2, \dots$ be independent with $\Pr(G_i = a) = 2^{-a}$, $a \geq 1$, and put

$$X = \sum_{j \geq 1} 3^{j-1} 2^{-(G_1 + \dots + G_j)} \quad \text{in } \mathbb{Z}_3, \quad \pi = \mathcal{L}(X). \quad (31)$$

The series converges 3-adically. Its projection modulo 3^k is Tao's Syracuse random variable [1, Section 1.4, equations (1.22)–(1.24)], after reversal of the independent valuation coordinates. The memoryless geometric law gives

$$\pi = \frac{1}{2} H_* \pi + \frac{1}{2} J_* \pi, \quad H(x) = x/2, \quad J(x) = (3x + 1)/2.$$

For $v \in \mathbb{Z}_3$, define

$$\rho_k(v) = 3^k \pi(v + 3^k \mathbb{Z}_3), \quad \rho_0(v) = 1. \quad (32)$$

The stationary equation gives, for $k \geq 1$,

$$\rho_k(v) = \frac{1}{2} \rho_k(2v) + \frac{3}{2} \mathbf{1}_{v \equiv 2 \pmod{3}} \rho_{k-1}((2v - 1)/3). \quad (33)$$

Indeed the inverse image under J is empty outside its unit class, and in that class a depth- k cylinder pulls back to a depth- $(k - 1)$ cylinder.

Iterating the first term, its remainder tends to zero since $\rho_k \leq 3^k$. Thus for positive integer v ,

$$\rho_k(v) = \mathcal{S}_1^k \mathbf{1}(v), \quad (\mathcal{S}_s f)(v) = \sum_{\substack{a \geq 1 \\ 2^a v \equiv 1 \pmod{3}}} (3/2^a)^s f((2^a v - 1)/3). \quad (34)$$

Every integral argument in this formula is a positive odd integer. An inverse word of depth k , with exponents of total $A = a_1 + \dots + a_k$, has weight $3^k 2^{-A}$. Read forward, it is a shortcut path with k odd steps and total length A . Distinct words with the same A have distinct endpoints, because an endpoint determines its forward path and hence its parity word.

Proposition 12. *For $k \geq 1$, let*

$$q_k = \max_{v \bmod 3^k} \pi(v + 3^k \mathbb{Z}_3).$$

Then

$$2^{-k} \leq q_k \leq (2k + 3)2^{-k}. \quad (35)$$

Consequently

$$\lim_{k \rightarrow \infty} \frac{-\log q_k}{k \log 3} = \log_3 2.$$

For every $\theta < \log_3 2$ there is C_θ such that $\pi(v + 3^k \mathbb{Z}_3) \leq C_\theta 3^{-\theta k}$ for all v, k ; no exponent larger than $\log_3 2$ has this property.

Proof. Choose a positive representative N of a cylinder. For fixed k, A , an admissible inverse word ends at an odd integer m with

$$T^A(m) = N = \frac{3^k}{2^A} m + b, \quad 0 \leq b \leq (3/2)^k - 1.$$

The offset bound follows by induction, since even steps divide the current offset and odd steps replace it by $3b/2 + 1/2$. The possible endpoints therefore lie in an interval of length strictly less than 2^{A-k} . Its odd integers number at most $2^{A-k-1} + 1$. By the fixed-length injectivity just noted, the contribution of this total A to the cylinder probability $3^{-k} \rho_k(N)$ is at most

$$2^{-A}(2^{A-k-1} + 1).$$

Summing $k \leq A < 5k$ gives at most $(2k + 2)2^{-k}$. For the remaining words, discard admissibility. Their total cylinder mass is at most

$$\Pr(G_1 + \dots + G_k \geq 5k) \leq (2/3)^{5k} (\mathbb{E}(3/2)^{G_1})^k = (32/81)^k \leq 2^{-k}.$$

This proves the upper bound.

On the event $G_1 = \dots = G_k = 1$, of probability 2^{-k} , the projection of (31) modulo 3^k is

$$\sum_{j=1}^k 3^{j-1} 2^{-j} = (3/2)^k - 1 \equiv -1 \pmod{3^k}.$$

This proves the lower bound. Taking logarithms gives the limiting exponent. The polynomial factor $2k + 3$ can be absorbed in $3^{(\log_3 2 - \theta)k}$ for every strictly smaller θ . The lower bound excludes any larger exponent. The bounds retain a possible polynomial loss at the endpoint exponent. $\square$

This estimate controls stationary cylinders uniformly. It does not provide an integer-height estimate uniform in the depth k . To control depth averages at fixed integers, we use the endpoint representation and the first-hit time law.

10 Identification of the cylinder and Green limits

For $0 < z < 1$, set

$$\mathcal{A}_z(N) = (1 - z) \sum_{k \geq 0} z^k \rho_k(N).$$

We first give the cutoff estimate that will control exceptional first-hit times.

Lemma 13. *The total weight of inverse odd-block words of depths $1 \leq k \leq K$ and endpoints greater than $N2^{6K}$ is at most*

$$\frac{9}{8}(64/81)^K. \quad (36)$$

Moreover $\rho_k(N) \leq 5kN + (64/81)^k$ for $k \geq 1$, so $\mathcal{A}_z(N)$ is finite.

Proof. For a depth- k word of total length A , its endpoint satisfies $m \leq 2^A N/3^k$. An endpoint greater than $N2^{6K}$ therefore requires $A > 6K$. Discarding admissibility and summing over compositions gives total weight at most

$$\sum_{k=1}^K 3^k \Pr(G_1 + \dots + G_k > 6K) \leq (2/3)^{6K} \sum_{k=1}^K 9^k \leq \frac{9}{8}(64/81)^K.$$

For a fixed depth k , apply the same Markov estimate to the complementary tail $A \geq 6k$; its bound is $(64/81)^k$. For each $k \leq A < 6k$, endpoints are distinct positive integers below $2^A N/3^k$, all with weight $3^k 2^{-A}$. Their combined weight is at most N . There are $5k$ such lengths, proving the remaining assertions. $\square$

Theorem 14. *For every positive integer N ,*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^K \rho_k(N) = \lim_{z \uparrow 1} \mathcal{A}_z(N) = \lim_{s \downarrow 1} C_s(N) = g(N) = \delta B_N N D_N. \quad (37)$$

No convergence of the individual sequence $\rho_k(N)$ is asserted.

Proof. The inverse odd-block expansion has a precise interpretation in terms of odd starting endpoints. A first-hit path from odd q to N has depth $k_N(q)$ and weight $Nw_N(q)/q$. For an even target the last block stops at N without necessarily continuing to its odd part; its exponent still counts exactly the shortcut steps in that block. Thus the representation remains exact.

Define the first-hit partial sum

$$F_N(K) = N \sum_{\substack{q \text{ odd}, q \in \mathcal{B}_N \\ k_N(q) \leq K}} \frac{w_N(q)}{q}.$$

For a nonperiodic target its terms are distinct inverse endpoints. For a periodic target it still selects only the first-hit word of each endpoint. In both cases Lemma 13 bounds its part with $q > N2^{6K}$ by (36). Within that cutoff the exceptional set in Proposition 10 contributes $o(K)$, since its reciprocal sum is $o(\log(N2^{6K}))$ and $0 \leq w_N \leq 1$.

Outside that exceptional set, for $0 < \epsilon < 1$,

$$(1 - \epsilon) \frac{\log q}{\lambda} \leq k_N(q) \leq (1 + \epsilon) \frac{\log q}{\lambda}.$$

Hence all good basin endpoints below $\exp(\lambda K/(1 + \epsilon))$ are counted, while every good counted endpoint is below $\exp(\lambda K/(1 - \epsilon))$. By Proposition 7,

$$\sum_{\substack{q \leq X \\ q \text{ odd}}} \frac{w_N(q)}{q} = \frac{D_N}{2} \log X + o(\log X).$$

The cutoff, the $o(K)$ exceptional contribution, and this identity give

$$\frac{\delta N D_N}{1 + \epsilon} \leq \liminf_{K \rightarrow \infty} \frac{F_N(K)}{K} \leq \limsup_{K \rightarrow \infty} \frac{F_N(K)}{K} \leq \frac{\delta N D_N}{1 - \epsilon}.$$

Letting $\epsilon \downarrow 0$ proves

$$F_N(K)/K \longrightarrow \delta N D_N. \quad (38)$$

Independently of the time law, the same cutoff and $w_N \leq 1$ give $F_N(K) \leq C_N(K+1)$.

Define $e_N = \mathbf{1}_{2|N}$. If N is nonperiodic, every odd endpoint hits it only once, and hence

$$\sum_{k=0}^K \rho_k(N) = e_N + F_N(K).$$

For odd N , the empty word is already counted by the endpoint $q = N$ in F_N . For even N , no odd endpoint has depth zero, so e_N supplies the single term $\rho_0(N) = 1$. If N is periodic, later visits repeat its cycle. Each odd endpoint contributes at depths $k_N(q) + r_N j$, with weights multiplied by β_N^j . Thus

$$\sum_{k=0}^K \rho_k(N) = e_N + \sum_{j=0}^{\lfloor K/r_N \rfloor} \beta_N^j F_N(K - r_N j).$$

The even depth-zero singleton is outside this sum: positive-depth odd-block words always have odd endpoints. The domination $F_N(K) \leq C_N(K+1)$ permits dominated convergence against the geometric series after division by K . Equation (38) gives the Cesàro limit $\delta N D_N / (1 - \beta_N)$.

In either case, summation by parts gives the elementary Abelian implication from a convergent Cesàro mean to $\mathcal{A}_z(N) \rightarrow g(N)$. The value agrees with Theorem 8, proving all of (37). $\square$

There is a useful check on the normalization. Every finite inverse word is uniquely partitioned, in traversal order, as

$$D^{a_1-1} F \dots D^{a_k-1} F D^j, \quad a_i \geq 1, \quad j \geq 0,$$

where $D(v) = 2v$ and $F(v) = (2v-1)/3$ is used only when integral. The last run contributes 2^{-sj} . Summing these nonnegative finite-word weights by Tonelli, and then the geometric series in j , yields

$$M_s(N) = \frac{1}{1-2^{-s}} \sum_{k \geq 0} \mathcal{S}_s^k \mathbf{1}(N).$$

With $\kappa'_s = 3^{s-1}/(2^s-1)$, one has

$$C_s(N) = (1 - \kappa'_s) \sum_{k \geq 0} \mathcal{S}_s^k \mathbf{1}(N), \quad 1 - \kappa'_s = \lambda(s-1) + O((s-1)^2).$$

This agrees with (37): odd endpoints have harmonic mean $D_N/2$, and $\lambda D_N/2 = \delta D_N$. No limit at a moving integer target was interchanged with a critical parameter limit.

11 Forward component mass and the remaining global condition

A tail component is called divergent if its orbits never enter a cycle; otherwise it is the basin of a finite cycle. The next proposition makes the remaining global estimate precise.

Proposition 15. *Let $\mathcal{C}$ be a divergent component and $n_j = T^j(n)$ any forward orbit in it. Then*

$$d_{\log}(\mathcal{B}_{n_j}) \longrightarrow \mu_{\text{tail}}(\mathcal{C}).$$

The normalized trace has a component-dependent limit $L_{\mathcal{C}}$ with

$$\frac{\delta}{P_0} \mu_{\text{tail}}(\mathcal{C}) \leq L_{\mathcal{C}} := \lim_{j \rightarrow \infty} \frac{g(n_j)}{n_j} \leq \delta \mu_{\text{tail}}(\mathcal{C}). \quad (39)$$

The following conditions are equivalent:

1. $g(v) = o(v)$ as $v \rightarrow \infty$ through nonperiodic targets;
2. every divergent component has μ_{tail} -mass zero;
3. the eventually-periodic set has logarithmic density one;
4. g vanishes on every divergent component.

Proof. The basins $\mathcal{B}_{n_j}$ increase and exhaust $\mathcal{C}$, but this alone would not justify continuity of logarithmic density. Fix M . Each of the finitely many $u \leq M$ in $\mathcal{C}$ meets the chosen spine. Thus, for some finite $J(M)$, every such u reaches n_j for $j \geq J(M)$. It follows that

$$\mathcal{C} \setminus \mathcal{B}_{n_j} \subseteq \{q : \min_{i \geq 0} T^i(q) > M\} \quad (j \geq J(M)).$$

Equation (16) and the existence of the relevant densities prove the asserted basin-density limit.

For $q \in \mathcal{C}$, Lemma 1 gives a convergent whole-orbit product

$$P_\infty(q) = \prod_{\substack{i \geq 0 \\ T^i(q) \text{ odd}}} \left(1 + \frac{1}{3T^i(q)}\right), \quad 1 \leq P_\infty(q) \leq P_0.$$

Set $W_{\mathcal{C}}(q) = 1/P_\infty(q)$ on $\mathcal{C}$ and zero elsewhere. On $\mathcal{B}_{n_j}$, first-hit factorization gives

$$W_{\mathcal{C}}(q) = \frac{w_{n_j}(q)}{P_\infty(n_j)}, \quad P_\infty(n_j) \rightarrow 1.$$

The observables $W_{\mathcal{C}} \mathbf{1}_{\mathcal{B}_{n_j}}$ increase, and their logarithmic means are $D_{n_j}/P_\infty(n_j)$. Their error from $W_{\mathcal{C}}$ is bounded by $\mathbf{1}_{\mathcal{C} \setminus \mathcal{B}_{n_j}}$, whose density tends to zero. Consequently the full weighted mean exists:

$$d_{\mathcal{C}} = \lim_{X \rightarrow \infty} \frac{1}{\log X} \sum_{\substack{q \leq X \\ q \in \mathcal{C}}} \frac{1}{q P_\infty(q)} = \lim_{j \rightarrow \infty} D_{n_j}.$$

Formula (22) gives $L_{\mathcal{C}} = \delta d_{\mathcal{C}}$, and the bounds on P_∞ prove (39).

Condition 1 implies 2 by applying (39) along each divergent spine. Conditions 2 and 3 are equivalent by countable additivity of μ_{tail} : bounded integer orbits are exactly the eventually periodic ones, and they form the union of the periodic components.

Assume condition 2 and fix $\epsilon > 0$. Choose finitely many periodic components $\mathcal{C}_1, \dots, \mathcal{C}_r$ whose total mass exceeds $1 - \epsilon$. For a cycle point c_i in each one, $\mathcal{C}_i = \mathcal{B}_{c_i}$. The uniform fixed-basin limit (26) makes $g(v)/v$ at most ϵ at all sufficiently large targets in their finite union. At a nonperiodic target outside that union, its basin is contained in the complement, so

$$\frac{g(v)}{v} \leq \delta d_{\log}(\mathcal{B}_v) \leq \delta \mu_{\text{tail}}(\mathcal{C}_1^c \cap \dots \cap \mathcal{C}_r^c) \leq \delta \epsilon.$$

This proves condition 1. Finally condition 2 forces density zero for every basin in a divergent component, hence $g = 0$ there by (22). Conversely a positive-mass divergent component has positive limiting ratio in (39), so cannot satisfy condition 4. $\square$

The restriction to nonperiodic targets avoids an unproved bound on the cycle factors $(1 - \beta_N)^{-1}$ across hypothetical additional cycles. The equivalent global condition in Proposition 15 has not been proved. The preceding Tao-based argument alone does not exclude nonempty components of logarithmic density zero. The additional predecessor input in the next section excludes that possibility and proves positivity at every unit. It still does not prove the global condition or rule out additional positive cycles.

12 Consequences of Mazur's natural-counting and predecessor theorems

This section uses two additional results of Mazur. His comparison of natural and logarithmic first-passage laws upgrades the density limits above to natural density. His positive lower bound for each eligible predecessor set determines the support of the canonical trace. The conclusions below are deductions from those inputs and the preceding sections; the inputs are not attributed to Tao's published theorem.

Put $\alpha = 1001/1000$. For the odd block

$$B_y = \{q \geq 1 : q \text{ odd}, y \leq q \leq y^\alpha\},$$

let U_y be its uniform law and L_y its law proportional to $1/q$. We use the following consequence of [8, Theorem 3.1]. For every $0 < d < 5/143$, all sufficiently large x , and either $y = x^\alpha$ or $y = x^{\alpha^2}$, the block is nonempty and

$$\Pr(\tau_x(U_y) = \infty) \leq 184x^{-1/32000}, \quad (40)$$

$$\|\mathcal{L}(\text{Pass}_x(U_y)) - \mathcal{L}(\text{Pass}_x(L_y))\|_1 \leq 384256(\log x)^{-d}. \quad (41)$$

Here $\|\cdot\|_1$ is full ℓ^1 distance, and first passage has the same default value one as above. The cited theorem actually bounds failure to hit by a specified finite time; the never-hit bound (40) is weaker.

Theorem 16. *Every label covered by Proposition 5 has the same limiting law under uniformly chosen odd starts in $[1, X]$ as under logarithmically chosen odd starts. Convergence is in total variation. Extending the label by taking the odd part gives the same limit under uniformly chosen positive starts.*

Every fixed shortcut basin $\mathcal{B}_N$ has natural density, and

$$d_{\text{nat}}(\mathcal{B}_N) = d_{\log}(\mathcal{B}_N), \quad \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{q \leq X} w_N(q) = D_N.$$

In particular, every union of tail components has natural density equal to its logarithmic density. The natural laws of orbit minima and of every fixed first-passage location also converge in total variation to the probabilities already constructed above.

Proof. Let F be one of the labels in Proposition 5, with values in a countable set I , and let μ be its limiting probability. In $\ell^1(I)$, put

$$A_F(t) = \sum_{\substack{q \leq e^t \\ q \text{ odd}}} \frac{\delta_F(q)}{q}.$$

The proposition gives $A_F(t)/t \rightarrow \mu/2$ in norm. Hence

$$\mathcal{L}(F(L_y)) = \frac{A_F(\alpha \log y) - A_F(\log y) + O_{\ell^1}(y^{-1})}{\frac{1}{2}(\alpha - 1) \log y + O(y^{-1})} \rightarrow \mu.$$

The endpoint error accounts for a possible included lower endpoint.

On successful passage the label is unchanged. Coupling a source with its own passage value therefore bounds the full ℓ^1 distance between its source-label and passage-label laws by twice its failure probability. Push (41) forward under F , and use (40) and Proposition 2 for the two failure terms. It follows that

$$\|\mathcal{L}(F(U_y)) - \mathcal{L}(F(L_y))\|_1 = o(1) \quad (y = x^\alpha, x \rightarrow \infty).$$

Thus $\mathcal{L}(F(U_y)) \rightarrow \mu$.

For an arbitrary large endpoint X , take

$$x = X^{1/\alpha^2}, \quad y = X^{1/\alpha}.$$

Then B_y is precisely the odd part of $[y, X]$. The proportion of positive odds below y among those at most X is $O(X^{1/\alpha-1}) = o(1)$. Replacing the uniform law on this block by the uniform law on all positive odds at most X changes full ℓ^1 distance by at most twice that proportion. This proves the odd-label assertion.

For the basin and weighted assertions, fix N . The only additional obstruction to passage invariance of $\mathbf{1}_{\mathcal{B}_N}$ is a skipped crossing of an even target. For $x \geq N^2$, that event is contained in $\text{Pass}_x \leq \sqrt{x}$, as in the proof of Proposition 6. Lemma 4 gives probability $O(x^{-c})$ under L_y . Transporting this event through (41) gives probability $o(1)$ under U_y as well. Failure to pass also has probability $o(1)$ for both sources.

On successful passage without a skipped target, the first-hit product factorization used in Proposition 7 gives

$$w_N(q) = \frac{w_N(\text{Pass}_x(q))}{P_{q \rightarrow \text{Pass}_x(q)}}.$$

Every odd source before the passage is greater than x , and the successful prefix has distinct states. The tail estimate (4) therefore implies

$$|w_N(q) - w_N(\text{Pass}_x(q))| = O(x^{b-1}) = o(1),$$

where the fixed packing exponent satisfies $b < 1$. Both observables lie in $[0, 1]$. Applying (41) to their passage values and including the preceding exceptional probabilities proves

$$\mathbb{E}_{U_y} f - \mathbb{E}_{L_y} f = o(1), \quad f = \mathbf{1}_{\mathcal{B}_N} \text{ or } w_N.$$

The logarithmic block means tend respectively to $d_{\log}(\mathcal{B}_N)$ and D_N , by Propositions 6 and 7, including their odd-relative normalizations. The same top-block argument therefore proves these limits for uniform odd starts.

It remains to pass to all positive starts. For labels extended by the odd part, doubling leaves the label unchanged. For the two scalar observables, doubling has the same property except when the initial halving chain hits N . All exceptional starts lie on the ray $\{2^j N : j \geq 0\}$, which has only $O_N(\log X)$ members below X . For each fixed h , odd starts multiplied by 2^h have asymptotic proportion 2^{-h-1} among all positive starts, and their normalized laws or means have the limits just proved. Sum over $0 \leq h \leq H$. The remaining starts are multiples of 2^{H+1} , with proportion at most $2^{-H-1} + O(X^{-1})$. First let $X \rightarrow \infty$, then $H \rightarrow \infty$. This proves the all-positive assertions and, for labels, preserves convergence in full ℓ^1 . $\square$

The next conclusion uses Mazur's separate predecessor theorem [9, Theorem 1.1]. Write C for the ordinary Collatz map, which sends an odd q to $3q + 1$, and let

$$\mathcal{P}(a) = \{q \geq 1 : C^j(q) = a \text{ for some } j \geq 0\}.$$

The cited theorem states that, for every fixed $a > 0$ with $3 \nmid a$, there are $c_a > 0$ and X_a such that

$$\#\{q < X : q \in \mathcal{P}(a)\} \geq c_a X \quad (X \geq X_a, X \in \mathbb{N}).$$

It assumes neither that the target reaches one nor that the target is nonperiodic. Constants may depend on the target.

Theorem 17. *For every positive integer N ,*

$$d_{\log}(\mathcal{B}_N) > 0 \iff D_N > 0 \iff g(N) > 0 \iff 3 \nmid N.$$

The same equivalence holds with natural density in place of logarithmic density. Every tail component has positive logarithmic and natural density. Consequently the equivalent global conditions in Proposition 15 are also equivalent to every positive orbit being eventually periodic.

Proof. The map comparison [9, equation (1.3)] is

$$\mathcal{P}(2N) \subseteq \mathcal{B}_N \subseteq \mathcal{P}(N).$$

For completeness, a shortcut orbit retains the ordinary orbit except for the intermediate even output immediately after an odd step. If an ordinary occurrence of $2N$ is retained, its next shortcut step reaches N . If it is omitted, that combined shortcut step itself ends at N . This proves the first inclusion; expanding shortcut steps proves the second.

If $3 \nmid N$, apply Mazur's theorem to $2N$. The first inclusion gives positive lower natural density of $\mathcal{B}_N$. It also gives positive lower logarithmic density: writing $A(t) = \#\{q \leq t : q \in \mathcal{B}_N\}$, the count bound gives $A(t) \geq c_{2N}t - O(1)$ for large real t , and summation by parts yields

$$\sum_{\substack{q \leq X \\ q \in \mathcal{B}_N}} \frac{1}{q} = \frac{A(X)}{X} + \int_1^X \frac{A(t)}{t^2} dt \geq c_{2N} \log X - O_N(1).$$

Thus the existing logarithmic density is positive, independently of the natural-density upgrade. The bounds (17) and the Green formula (22) give $D_N > 0$ and $g(N) > 0$.

If $3 \mid N$, the only shortcut predecessor of any positive multiple of three is its double: an odd shortcut step has output congruent to two modulo three. Inducting backward along a finite path gives

$$\mathcal{B}_N = \{2^j N : j \geq 0\}.$$

Its natural and logarithmic densities are zero. Hence $D_N = 0$ and $g(N) = 0$, again by the weighted bounds and Green formula. Theorem 16 supplies the asserted natural-density equivalence.

Every positive orbit reaches an integer prime to three. Indeed, remove its initial powers of two; an odd multiple of three has its next shortcut value prime to three. A tail component containing such a value N also contains $\mathcal{B}_N$, so its logarithmic density is positive. Its natural density is the same by Theorem 16.

One of the equivalent conditions in Proposition 15 is that every divergent component have zero logarithmic density. Since every component has positive density, this now means that there are no divergent components. That is exactly eventual periodicity of all positive orbits. Conversely, if every orbit is eventually periodic, the eventually periodic set is all positive integers, so the conditions of that proposition hold. $\square$

The positivity of the trace on units thus follows after adding Mazur's predecessor theorem. Global sublinearity remains unproved; with this additional input it is equivalent to excluding divergent orbits. Even that conclusion would still permit cycles other than the known cycle through one. Excluding such cycles, and hence proving the Collatz conjecture, is not achieved here.

13 Formal verification scope

The accompanying Lean 4.30.0-rc2 development gives checked declarations for the numbered results in Sections 2–12. It includes the Banach-valued two-scale averaging theorem and rate,

uniform trajectory packing and correction products, both real-threshold first-passage scales, local clock and atom estimates, and the countable label laws. The actual weighted and ordinary basin means are proved from those inputs. The clock-exception estimates are then proved for odd and all positive starts, and the Green, cylinder depth, Cesàro, and Abel limits are identified at each positive target without additional density or clock hypotheses.

The native cylinder development constructs the canonical 3-adic probability law and checks stationarity, uniqueness, the explicit recursion, maximal-cylinder bounds and exponent, and the inverse-block normalizations. It also proves the uniform backward-basin boundary estimate, forward-component exhaustion and weighted limits, and the four global-condition equivalences. Mazur’s natural-counting and predecessor theorems are included with their actual map and counting interfaces. They yield the natural-density limits, positivity exactly at targets prime to three, positive component masses, and the stated equivalence with universal eventual periodicity. The equivalent global conditions remain unproved; the development does not exclude additional cycles or prove the Collatz conjecture.

The scales 1001/1000 and 2001/2000 use distinct namespaces in one Lean environment, with their numerical definitions checked by the root audit. Lemma 4 is stated at those two scales. The predecessor source has a third namespace because its pinned baseline differs from the first-passage package. Imported sources retain their original attribution and licenses.

The combined bundle contains 1,602 dependency modules, including 281 local proof modules. Its root audit covers 1,507 local theorem and lemma declarations, 275 definitions and named instances, and 5 upstream theorem roots. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted in their logical dependencies. The source manifest, dependency pins, theorem map, and recorded build are supplied in the repository. The recorded joint build identifies fresh compilation and exact dependency-checked reuse; the default verifier rebuilds all bundled modules. The earlier Lean 4.27 development is retained separately for reproducibility. The exact correspondence between manuscript statements and declarations is recorded in `paper/basin-lean-claims.md`.

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Use of AI

OpenAI Codex was used extensively for mathematical exploration and preparation and revision of this manuscript. The proofs use the explicitly cited analytic inputs. The numbered results have accompanying Lean proofs; their source and verification boundaries are specified above. Automated review does not constitute external peer review or endorsement by the authors of the cited works.

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